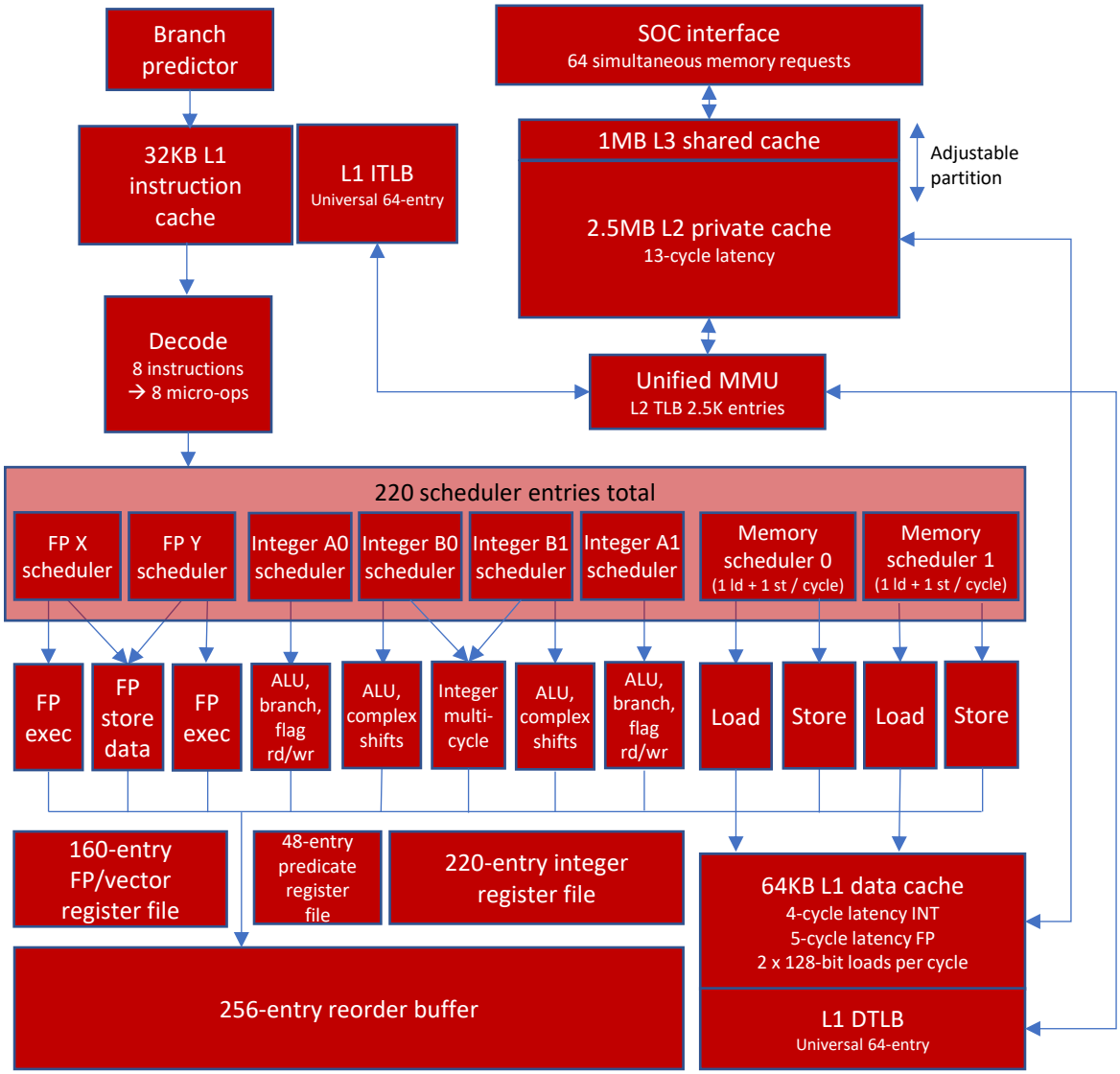

Branch predictor
SOC interface
64 simultaneous memory requests
32KB L1 instruction cache
L1 ITLB
Universal 64-entry
1MB L3 shared cache
Adjustable partition
2.5MB L2 private cache
13-cycle latency
Decode
8 instructions
→ 8 micro-ops
Unified MMU
L2 TLB 2.5K entries
220 scheduler entries total
FP X scheduler
FP Y scheduler
Integer A0 scheduler
Integer B0 scheduler
Integer B1 scheduler
Integer A1 scheduler
Memory scheduler 0
(1 ld + 1 st / cycle)
Memory scheduler 1
(1 ld + 1 st / cycle)
FP exec
FP store data
FP exec
ALU, branch, flag rd/wr
ALU, complex shifts
Integer multi-cycle
ALU, complex shifts
ALU, branch, flag rd/wr
Load
Store
Load
Store
160-entry FP/vector register file
48-entry predicate register file
220-entry integer register file
64KB L1 data cache
4-cycle latency INT
5-cycle latency FP
2 x 128-bit loads per cycle
256-entry reorder buffer
L1 DTLB
Universal 64-entry